

Database Schema Documentation

Overview

This document describes the complete database schema for the Vidyarthi Book Depot platform. The database uses MySQL 8.0+ with InnoDB storage engine and UTF-8 character set.

Database Configuration

- **Database Name:** vidyarthi_book_depot
- **Character Set:** utf8mb4
- **Collation:** utf8mb4_unicode_ci
- **Storage Engine:** InnoDB
- **Time Zone:** UTC

Entity Relationship Diagram

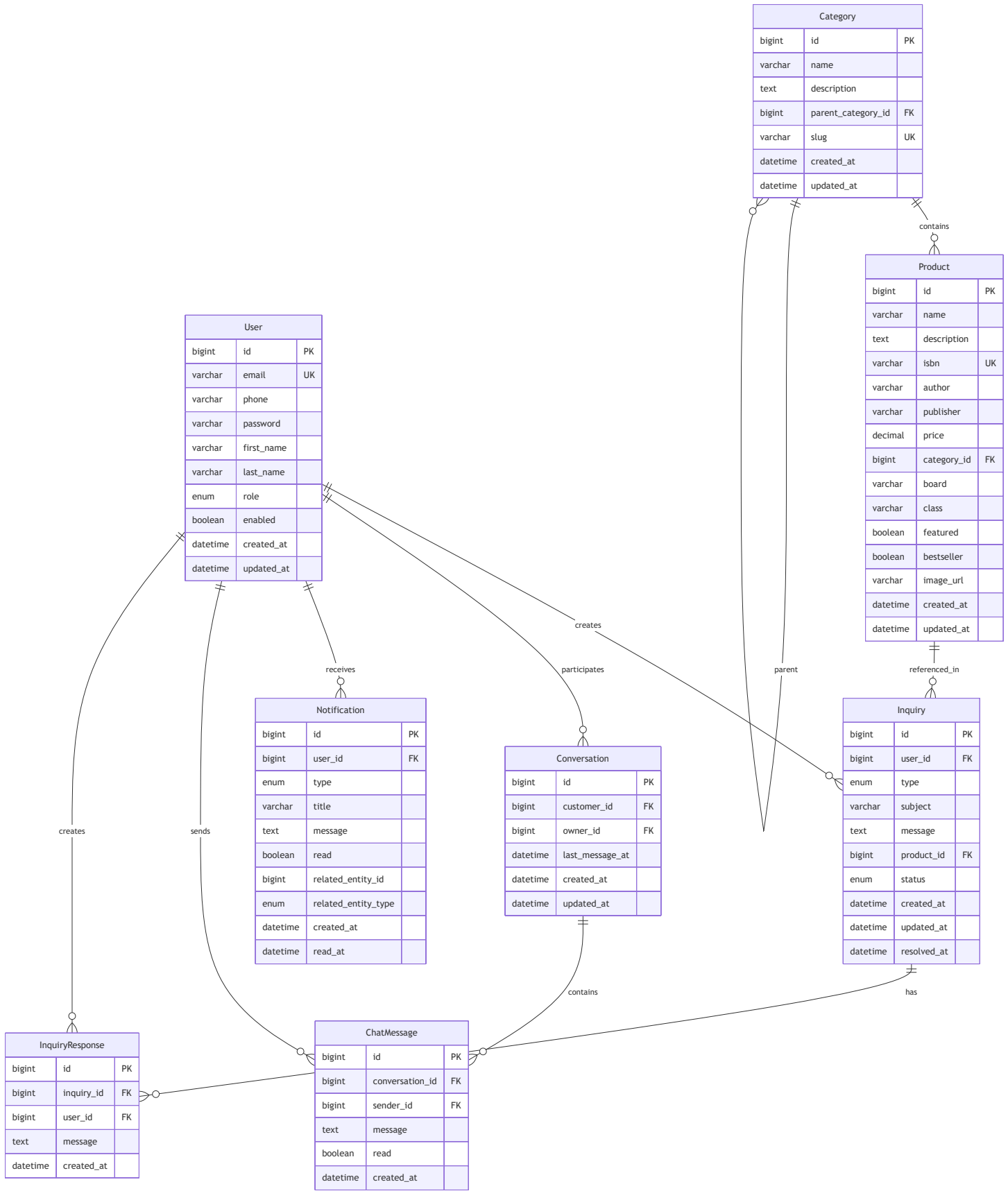


Table Definitions

1. users

Description: Stores user account information

Columns:

Column Name	Data Type	Constraints	Description
id	BIGINT	PRIMARY KEY, AUTO_INCREMENT	User ID
email	VARCHAR(255)	UNIQUE, NOT NULL	User email address
phone	VARCHAR(20)	NULL	Phone number
password	VARCHAR(255)	NOT NULL	Encrypted password (BCrypt)
first_name	VARCHAR(100)	NULL	First name
last_name	VARCHAR(100)	NULL	Last name
role	ENUM('CUSTOMER', 'OWNER', 'ADMIN')	NOT NULL, DEFAULT 'CUSTOMER'	User role
enabled	BOOLEAN	NOT NULL, DEFAULT TRUE	Account enabled status
created_at	DATETIME	NOT NULL	Creation timestamp
updated_at	DATETIME	NOT NULL	Last update timestamp

Indexes:

- PRIMARY KEY (id)
- UNIQUE KEY (email)

- INDEX (role)
- INDEX (enabled)

Sample Data:

```
INSERT INTO users (email, phone, password, first_name, last_name, role, enabled)
VALUES
('owner@vbd.com', '+919876543210', '$2a$10$...', 'Store', 'Owner', 'OWNER', TRUE),
('customer@example.com', '+919876543211', '$2a$10$...', 'John', 'Doe', 'CUSTOMER', TRUE);
```

2. products

Description: Stores product/book information

Columns:

Column Name	Data Type	Constraints	Description
id	BIGINT	PRIMARY KEY, AUTO_INCREMENT	Product ID
name	VARCHAR(255)	NOT NULL	Product name
description	TEXT	NULL	Product description
isbn	VARCHAR(50)	UNIQUE	ISBN number
author	VARCHAR(255)	NULL	Author name
publisher	VARCHAR(255)	NULL	Publisher name
price	DECIMAL(10,2)	NOT NULL	Product price
category_id	BIGINT	FOREIGN KEY	Category ID
board	VARCHAR(50)	NULL	Educational board (SSC, HSC, ICSE, CBSE)
class	VARCHAR(50)	NULL	Class/grade
featured	BOOLEAN	NOT NULL, DEFAULT FALSE	Featured product flag

Column Name	Data Type	Constraints	Description
bestseller	BOOLEAN	NOT NULL, DEFAULT FALSE	Bestseller flag
image_url	VARCHAR(500)	NULL	Product image URL
created_at	DATETIME	NOT NULL	Creation timestamp
updated_at	DATETIME	NOT NULL	Last update timestamp

Indexes:

- PRIMARY KEY (id)
- UNIQUE KEY (isbn)
- INDEX (category_id)
- INDEX (board)
- INDEX (class)
- INDEX (featured)
- INDEX (bestseller)
- FULLTEXT INDEX (name, description) -- For search

Foreign Keys:

- FOREIGN KEY (category_id) REFERENCES categories(id) ON DELETE SET NULL

Sample Data:

```
INSERT INTO products (name, description, isbn, author, publisher, price, category_id, board, class)
VALUES
('Mathematics for Class 10', 'Comprehensive mathematics textbook', '978-1234567890', 'Dr. Ramesh', 'ABC Publications', 120, 1, 'CBSE', '10th Grade')
```

3. categories

Description: Stores product categories with hierarchical structure

Columns:

Column Name	Data Type	Constraints	Description
id	BIGINT	PRIMARY KEY, AUTO_INCREMENT	Category ID
name	VARCHAR(255)	NOT NULL	Category name
description	TEXT	NULL	Category description
parent_category_id	BIGINT	FOREIGN KEY, NULL	Parent category ID
slug	VARCHAR(255)	UNIQUE	URL-friendly slug
created_at	DATETIME	NOT NULL	Creation timestamp
updated_at	DATETIME	NOT NULL	Last update timestamp

Indexes:

- PRIMARY KEY (id)
- UNIQUE KEY (slug)
- INDEX (parent_category_id)

Foreign Keys:

- FOREIGN KEY (parent_category_id) REFERENCES categories(id) ON DELETE CASCADE

Sample Data:

```
INSERT INTO categories (name, description, parent_category_id, slug)
VALUES
('School Books', 'Textbooks for school students', NULL, 'school-books'),
('SSC', 'State Board books', 1, 'school-books-ssc'),
('HSC', 'Higher Secondary books', 1, 'school-books-hsc');
```

4. inquiries

Description: Stores customer inquiries

Columns:

Column Name	Data Type	Constraints	Description
id	BIGINT	PRIMARY KEY, AUTO_INCREMENT	Inquiry ID
user_id	BIGINT	FOREIGN KEY, NOT NULL	User ID (customer)
type	ENUM('BOOK_AVAILABILITY', 'GENERAL_QUESTION', 'CUSTOM_ORDER')	NOT NULL	Inquiry type
subject	VARCHAR(200)	NOT NULL	Inquiry subject
message	TEXT	NOT NULL	Inquiry message
product_id	BIGINT	FOREIGN KEY, NULL	Related product ID
status	ENUM('PENDING', 'IN_PROGRESS', 'RESOLVED', 'CLOSED')	NOT NULL, DEFAULT 'PENDING'	Inquiry status
created_at	DATETIME	NOT NULL	Creation timestamp
updated_at	DATETIME	NOT NULL	Last update timestamp
resolved_at	DATETIME	NULL	Resolution timestamp

Indexes:

- PRIMARY KEY (id)
- INDEX (user_id)
- INDEX (product_id)
- INDEX (status)
- INDEX (type)
- INDEX (created_at)

Foreign Keys:

- FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE
- FOREIGN KEY (product_id) REFERENCES products(id) ON DELETE SET NULL

Sample Data:

```
INSERT INTO inquiries (user_id, type, subject, message, product_id, status)
VALUES
(2, 'BOOK_AVAILABILITY', 'Availability of Mathematics Class 10', 'Do you have this book in stock', 1, 'PENDING')
```

5. inquiry_responses

Description: Stores owner responses to inquiries

Columns:

Column Name	Data Type	Constraints	Description
id	BIGINT	PRIMARY KEY, AUTO_INCREMENT	Response ID
inquiry_id	BIGINT	FOREIGN KEY, NOT NULL	Inquiry ID
user_id	BIGINT	FOREIGN KEY, NOT NULL	User ID (owner)
message	TEXT	NOT NULL	Response message
created_at	DATETIME	NOT NULL	Creation timestamp

Indexes:

- PRIMARY KEY (id)
- INDEX (inquiry_id)
- INDEX (user_id)
- INDEX (created_at)

Foreign Keys:

- FOREIGN KEY (inquiry_id) REFERENCES inquiries(id) ON DELETE CASCADE
- FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE

Sample Data:


```
INSERT INTO inquiry_responses (inquiry_id, user_id, message)
VALUES
(1, 1, 'Yes, we have this book in stock. You can visit our store.');
```

6. conversations

Description: Stores chat conversations between customers and owner

Columns:

Column Name	Data Type	Constraints	Description
id	BIGINT	PRIMARY KEY, AUTO_INCREMENT	Conversation ID
customer_id	BIGINT	FOREIGN KEY, NOT NULL	Customer user ID
owner_id	BIGINT	FOREIGN KEY, NOT NULL	Owner user ID
last_message_at	DATETIME	NULL	Last message timestamp
created_at	DATETIME	NOT NULL	Creation timestamp
updated_at	DATETIME	NOT NULL	Last update timestamp

Indexes:

- PRIMARY KEY (id)
- UNIQUE KEY (customer_id, owner_id) -- One conversation per customer-owner pair
- INDEX (customer_id)
- INDEX (owner_id)
- INDEX (last_message_at)

Foreign Keys:

- FOREIGN KEY (customer_id) REFERENCES users(id) ON DELETE CASCADE
- FOREIGN KEY (owner_id) REFERENCES users(id) ON DELETE CASCADE

Sample Data:

```
INSERT INTO conversations (customer_id, owner_id, last_message_at)
VALUES
(2, 1, '2025-12-23 10:00:00');
```

7. chat_messages

Description: Stores chat messages

Columns:

Column Name	Data Type	Constraints	Description
id	BIGINT	PRIMARY KEY, AUTO_INCREMENT	Message ID
conversation_id	BIGINT	FOREIGN KEY, NOT NULL	Conversation ID
sender_id	BIGINT	FOREIGN KEY, NOT NULL	Sender user ID
message	TEXT	NOT NULL	Message content
read	BOOLEAN	NOT NULL, DEFAULT FALSE	Read status
created_at	DATETIME	NOT NULL	Creation timestamp

Indexes:

- PRIMARY KEY (id)
- INDEX (conversation_id)
- INDEX (sender_id)
- INDEX (conversation_id, created_at) -- For message ordering
- INDEX (conversation_id, read) -- For unread count

Foreign Keys:

- FOREIGN KEY (conversation_id) REFERENCES conversations(id) ON DELETE CASCADE
- FOREIGN KEY (sender_id) REFERENCES users(id) ON DELETE CASCADE

Sample Data:

```
INSERT INTO chat_messages (conversation_id, sender_id, message, read)
VALUES
(1, 2, 'Hello, do you have Mathematics Class 10?', FALSE),
(1, 1, 'Yes, we have it in stock.', TRUE);
```

8. notifications

Description: Stores in-app notifications

Columns:

Column Name	Data Type	Constraints	Description
id	BIGINT	PRIMARY KEY, AUTO_INCREMENT	Notification ID
user_id	BIGINT	FOREIGN KEY, NOT NULL	User ID
type	ENUM('INQUIRY_RESPONSE', 'CHAT_MESSAGE', 'INQUIRY_CREATED', 'SYSTEM')	NOT NULL	Notification type
title	VARCHAR(255)	NOT NULL	Notification title
message	TEXT	NOT NULL	Notification message
read	BOOLEAN	NOT NULL, DEFAULT FALSE	Read status
related_entity_id	BIGINT	NULL	Related entity ID
related_entity_type	ENUM('INQUIRY', 'CONVERSATION', 'PRODUCT', 'USER')	NULL	Related entity type
created_at	DATETIME	NOT NULL	Creation timestamp
read_at	DATETIME	NULL	Read timestamp

Indexes:

- PRIMARY KEY (id)

- INDEX (user_id)
- INDEX (type)
- INDEX (read)
- INDEX (user_id, read, created_at) -- For notification queries
- INDEX (user_id, type) -- For type-based queries

Foreign Keys:

- FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE

Sample Data:

```
INSERT INTO notifications (user_id, type, title, message, related_entity_id, related_entity_type)
VALUES
(2, 'INQUIRY_RESPONSE', 'Response to your inquiry', 'Store owner has responded to your inquiry',
```

Database Initialization Script

Schema Creation

```
-- Create database
CREATE DATABASE IF NOT EXISTS vidyarthi_book_depot
CHARACTER SET utf8mb4
COLLATE utf8mb4_unicode_ci;

USE vidyarthi_book_depot;

-- Create users table
CREATE TABLE users (
    id BIGINT AUTO_INCREMENT PRIMARY KEY,
    email VARCHAR(255) UNIQUE NOT NULL,
    phone VARCHAR(20),
    password VARCHAR(255) NOT NULL,
    first_name VARCHAR(100),
    last_name VARCHAR(100),
    role ENUM('CUSTOMER', 'OWNER', 'ADMIN') NOT NULL DEFAULT 'CUSTOMER',
    enabled BOOLEAN NOT NULL DEFAULT TRUE,
    created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    INDEX idx_role (role),
    INDEX idx_enabled (enabled)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

-- Create categories table
CREATE TABLE categories (
    id BIGINT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(255) NOT NULL,
    description TEXT,
    parent_category_id BIGINT,
    slug VARCHAR(255) UNIQUE NOT NULL,
    created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    INDEX idx_parent (parent_category_id),
    FOREIGN KEY (parent_category_id) REFERENCES categories(id) ON DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

-- Create products table
CREATE TABLE products (
```

```

id BIGINT AUTO_INCREMENT PRIMARY KEY,
name VARCHAR(255) NOT NULL,
description TEXT,
isbn VARCHAR(50) UNIQUE,
author VARCHAR(255),
publisher VARCHAR(255),
price DECIMAL(10,2) NOT NULL,
category_id BIGINT,
board VARCHAR(50),
class VARCHAR(50),
featured BOOLEAN NOT NULL DEFAULT FALSE,
bestseller BOOLEAN NOT NULL DEFAULT FALSE,
image_url VARCHAR(500),
created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
INDEX idx_category (category_id),
INDEX idx_board (board),
INDEX idx_class (class),
INDEX idx_featured (featured),
INDEX idx_bestseller (bestseller),
FULLTEXT INDEX idx_search (name, description),
FOREIGN KEY (category_id) REFERENCES categories(id) ON DELETE SET NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

-- Create inquiries table
CREATE TABLE inquiries (
id BIGINT AUTO_INCREMENT PRIMARY KEY,
user_id BIGINT NOT NULL,
type ENUM('BOOK_AVAILABILITY', 'GENERAL_QUESTION', 'CUSTOM_ORDER') NOT NULL,
subject VARCHAR(200) NOT NULL,
message TEXT NOT NULL,
product_id BIGINT,
status ENUM('PENDING', 'IN_PROGRESS', 'RESOLVED', 'CLOSED') NOT NULL DEFAULT 'PENDING',
created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
resolved_at DATETIME,
INDEX idx_user (user_id),
INDEX idx_product (product_id),
INDEX idx_status (status),
INDEX idx_type (type),
INDEX idx_created (created_at),
FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE,
FOREIGN KEY (product_id) REFERENCES products(id) ON DELETE SET NULL

```

```
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

```
-- Create inquiry_responses table
```

```
CREATE TABLE inquiry_responses (  
    id BIGINT AUTO_INCREMENT PRIMARY KEY,  
    inquiry_id BIGINT NOT NULL,  
    user_id BIGINT NOT NULL,  
    message TEXT NOT NULL,  
    created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,  
    INDEX idx_inquiry (inquiry_id),  
    INDEX idx_user (user_id),  
    INDEX idx_created (created_at),  
    FOREIGN KEY (inquiry_id) REFERENCES inquiries(id) ON DELETE CASCADE,  
    FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

```
-- Create conversations table
```

```
CREATE TABLE conversations (  
    id BIGINT AUTO_INCREMENT PRIMARY KEY,  
    customer_id BIGINT NOT NULL,  
    owner_id BIGINT NOT NULL,  
    last_message_at DATETIME,  
    created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,  
    updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,  
    UNIQUE KEY uk_customer_owner (customer_id, owner_id),  
    INDEX idx_customer (customer_id),  
    INDEX idx_owner (owner_id),  
    INDEX idx_last_message (last_message_at),  
    FOREIGN KEY (customer_id) REFERENCES users(id) ON DELETE CASCADE,  
    FOREIGN KEY (owner_id) REFERENCES users(id) ON DELETE CASCADE  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

```
-- Create chat_messages table
```

```
CREATE TABLE chat_messages (  
    id BIGINT AUTO_INCREMENT PRIMARY KEY,  
    conversation_id BIGINT NOT NULL,  
    sender_id BIGINT NOT NULL,  
    message TEXT NOT NULL,  
    read BOOLEAN NOT NULL DEFAULT FALSE,  
    created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,  
    INDEX idx_conversation (conversation_id),  
    INDEX idx_sender (sender_id),  
    INDEX idx_conversation_created (conversation_id, created_at),
```

```

    INDEX idx_conversation_read (conversation_id, read),
    FOREIGN KEY (conversation_id) REFERENCES conversations(id) ON DELETE CASCADE,
    FOREIGN KEY (sender_id) REFERENCES users(id) ON DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

-- Create notifications table
CREATE TABLE notifications (
    id BIGINT AUTO_INCREMENT PRIMARY KEY,
    user_id BIGINT NOT NULL,
    type ENUM('INQUIRY_RESPONSE', 'CHAT_MESSAGE', 'INQUIRY_CREATED', 'SYSTEM') NOT NULL,
    title VARCHAR(255) NOT NULL,
    message TEXT NOT NULL,
    read BOOLEAN NOT NULL DEFAULT FALSE,
    related_entity_id BIGINT,
    related_entity_type ENUM('INQUIRY', 'CONVERSATION', 'PRODUCT', 'USER'),
    created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
    read_at DATETIME,
    INDEX idx_user (user_id),
    INDEX idx_type (type),
    INDEX idx_read (read),
    INDEX idx_user_read_created (user_id, read, created_at),
    INDEX idx_user_type (user_id, type),
    FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

```

Migration Strategy

Using Flyway or Liquibase

The database schema should be managed using migration tools:

1. **Initial Migration:** Create all tables
2. **Version Control:** Track schema changes
3. **Rollback Support:** Ability to rollback changes
4. **Data Migrations:** Handle data transformations

Performance Optimization

Indexing Strategy

- **Primary Keys:** All tables have auto-increment primary keys
- **Foreign Keys:** Indexed for join performance
- **Query Patterns:** Indexes based on common query patterns
- **Composite Indexes:** For multi-column queries

Query Optimization

- **JOIN Optimization:** Proper foreign key relationships
- **Pagination:** All list queries support pagination
- **Full-Text Search:** Full-text indexes for product search
- **Covering Indexes:** Indexes that cover query requirements

Backup and Recovery

Backup Strategy

- **Daily Backups:** Full database backup
- **Transaction Logs:** Continuous transaction log backup
- **Point-in-Time Recovery:** Ability to recover to specific time

Recovery Procedures

1. Restore from latest backup
2. Apply transaction logs
3. Verify data integrity
4. Resume operations

Data Retention

Retention Policies

- **Active Data:** No automatic deletion
- **Archived Data:** Archive old inquiries after 1 year

- **Notifications:** Delete read notifications after 90 days
- **Chat Messages:** Retain all messages

Security Considerations

1. **User Permissions:** Database user with minimal required permissions
2. **SQL Injection:** Use parameterized queries (JPA handles this)
3. **Data Encryption:** Sensitive data encrypted at application level
4. **Backup Encryption:** Encrypted backups
5. **Access Control:** Database access restricted to application servers